

Gunjan Singh

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EDUCATION

- Indira Gandhi Delhi Technical University for Women** Delhi, India
 - Bachelor of Technology - Electronics and Communication; GPA: 7.26* *July 2025-Present*
 - Courses: Analog Electronics, Digital Electronics, Signal and Systems, Analog System, Verilog, C++, OOps*

SKILLS SUMMARY

- Languages:** C++, Python, JavaScript, HTML/CSS
- Frameworks:** React, Next.js, Tailwind CSS, Pandas, NumPy
- Hardware:** Arduino, VLSI, SPI/I2C Protocols, ADC/PWM, EEPROM
- Tools & Core:** Tableau, Machine Learning, Data Structures & Algorithms, Git
- Soft Skills:** Leadership, Event Management, Public Speaking, Time Management

VOLUNTEER EXPERIENCE

- Electronix Amplified** Remote
 - Student Developer Head (Full-time)* *May 2026-Present*
 - Mentorship:** Directed academic mentorship programs for a cohort of 100+ ECE and ECE-AI undergraduates, optimizing exam preparation strategies for core engineering and advanced technical curriculums.
 - Workshops:** Facilitated targeted scholarship application workshops guiding peers through competitive STEM foundation grants and corporate funding pipelines to increase cohort selection rates.
 - Resource Management:** Streamlined resource distribution and structured peer-led tutoring schedules, driving technical community engagement and measurable improvements in academic performance metrics.

PROJECTS

- ElectronixAmplified** | *Next.js, Tailwind CSS, React, Web UI/UX*
 - Project Overview:** Developed a comprehensive 8-semester web portal tailored for the ECE curriculum, centralizing academic notes, PYQs, and technical study materials.
 - Architecture:** Architected a responsive user interface utilizing modern web frameworks to ensure seamless navigation and rapid resource retrieval for the university engineering community.
- ThermalSync: Closed-Loop Cooling System** | *C++, Arduino, ADC, PWM, EEPROM*
 - Engineered a responsive climate control system utilizing an NTC Thermistor and an NPN Transistor to dynamically adjust DC motor fan speeds based on real-time environmental data.
 - Implemented the Steinhart-Hart equation for precise analog-to-digital signal processing, translating raw voltage into highly accurate Celsius measurements.
 - Designed proportional control logic for smooth mechanical acceleration and integrated EEPROM storage to retain custom temperature thresholds across hardware resets.
 - Developed a dual-mode interactive GUI on a 16x2 LCD, allowing users to monitor live metrics and calibrate system limits via hardware interrupts.
- TUX Paint: Capacitive Touch Digital Canvas** | *C++, Arduino, SPI/I2C Protocols*
 - Developed a standalone digital drawing application by integrating an Adafruit ILI9341 TFT display with an FT6206 touch sensor using simultaneous SPI and I2C protocols.
 - Programmed a dynamic GUI utilizing the Adafruit GFX library, featuring a real-time drawing canvas and a responsive top-bar color selection palette.
 - Implemented mathematical coordinate mapping algorithms to translate raw capacitive hardware inputs into calibrated screen dimensions for high-precision digital brushstrokes.
 - Optimized the continuous polling loop to achieve near-zero latency between physical touch and pixel rendering, delivering a highly tactile and interactive user experience.

HONORS AND AWARDS

- Infosys Foundation STEM Stars Scholar - *Selected for competitive merit-based funding program (Nov 2025)*
- Katalyst India Scholar - *Recognized for academic excellence and technical potential*